2. Collection Framework

a. ArrayList and LinkedList
b. Iterator and Iterable
c. Comparator and Comparable
d. HashMap and TreeMap
e. HashSet and TreeSet
f. HashTable

UNIT – V

Exception Handling: Introduction, Keywords: try, catch, throw, throws and finally, Multiple Catch Clauses, Checked and Unchecked Exceptions, User Defined Exceptions.

Multithreading: Introduction, Thread Life Cycle, Creation, Priorities, Synchronization, ITC.

Java Database Connectivity: Introduction, JDBC Architecture, Types of Drivers, JDBC Environment Setup, JDBC Steps, CRUD Operations.

Practice:

1. Exception Handling

a. Illustrate exception handling mechanism using multiple catch clauses.
b. Make use of Built-in and user-defined Exceptions in handling a run time exception.

2. Multithreading

a. Creates threads by extending Thread class. First thread display "Good Morning "every 1 sec, the second thread displays "Hello "every 2 seconds and the third display "Welcome" every 3 seconds, (Repeat the same by implementing Runnable).
b. Solve Producer-Consumer problem using synchronization.

3. JDBC

a. CRUD operations.

Capstone Project:

PROJECT 1: Student Attendance Management System

Project Implementation, Documentation and Viva Voce

- **Implementation mainly focuses on**
  - Adding student details, marking and viewing attendance.
  - JDBC operations: insert, update, search by roll number.
  - Input validation (no duplicate roll numbers, days ≤ total days).
  - Attendance percentage calculation.
  - Menu-driven console workflow with exception handling.
- **Documentation must contain**
  - Objective & problem definition
  - Attendance table structure
  - ER diagram (Student → Attendance)
  - Module explanations: Add, Mark, View, Search
  - JDBC connection flow
  - Screenshots of sample input/output
  - Enhancements added (login, validation, normalization)

PROJECT 2: Library Management System

- **Implementation mainly focuses on**
  - Adding, searching, updating, deleting book records.
  - JDBC CRUD operations on books table.
  - Validation of price, quantity.
  - Implementing search by ID, title, and author.

- **Documentation must contain**
  - Objective of library automation
  - Books table schema
  - Flowchart for Add → View → Delete operations
  - Module descriptions
  - Exception handling (invalid ID)
  - Screenshots
  - Enhancements (issue records, author search)

PROJECT 3: Student Course Registration System

Implementation Mainly Focuses On

- Course registration, view, update, delete operations.
- Database design with student, course, registration tables.
- Validation for duplicate registration.
- JDBC connectivity and table relationships.

Documentation Must Contain

- Objective
- Table design (student, course, registration)
- ER diagram
- Module explanation: Register, Display, Edit
- SQL constraints used
- Screenshots of registration process

PROJECT 4: Online Student Result Management System

Implementation Mainly Focuses On

- Marks entry and validation.
- Automatic total, percentage, grade calculation.
- Result storage and retrieval using JDBC.
- Web-based form input using HTML.

Documentation Must Contain

- Objective
- Marks table schema
- Grade calculation logic
- HTML form screenshots
- JDBC code explanation
- Enhancements (validation, grade generation)

PROJECT 5: Employee Leave Management System

Implementation Mainly Focuses On

- Leave application submission.
- Balance update logic.
- Validations on leave days and employee ID.
- JDBC-based leave approval or rejection.

Documentation Must Contain

- Objective
- Employee & leave tables
- Flowchart: Apply → Validate → Approve
- Module explanation
- Screenshots of leave form and output

PROJECT 6: Online Shopping Cart Simulation

Implementation Mainly Focuses On

- Displaying products and adding items to cart.
- Stock validation before adding to cart.
- Bill calculation with discounts/coupons.
- JDBC for storing cart and order history.

Documentation Must Contain

- Objective
- Product, cart, order tables
- Module flow: Add → Update → Bill
- Billing logic with formulas
- Screenshots of cart process

PROJECT 7: Bus Ticket Booking System

Implementation Mainly Focuses On

- Booking, cancelling, viewing seat availability.
- Preventing double booking of same seat.
- Route table and seat allocation logic.
- JDBC insert/update operations.

Documentation Must Contain

- Objective
- Route and seat tables
- Seat allocation flowchart
- Module descriptions
- SQL queries
- Screenshots of booking/cancel pages

Text Books:

1. The Complete Reference Java, Herbert Schildt, TMH ,8th Edition, ISBN 978-1260440232.
2. Java one step ahead, Anita seth, B.L.Juneja, Oxford, First Edition. ISBN: 9780199459643.

Reference Books:

1. Introduction to java programming, by Y Daniel Liang, Seventh Edition, Pearson, ISBN: 978-8131729588
2. Core Java: An Integrated Approach, R.Nageswara Rao, Dream tech press, ISBN: 978-9351199250